

24/8/24

classmate

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State of material Classification of GMPs / RP based on the raw material.

State of Material	Type	Mechanism	Energy Type	Energy Source	Process
Liquid	Photo-polymer	Liquid photo polymerization	Monochromatic light	Lamp Laser beam Holography	solid ground curing stereolithography Holographic interference solid
	Thermo	Liquid thermal	Light (two frequencies)	Two laser beams	Beam interference solidification
	Thermo	Liquid thermal	Heat	Laser beam	Thermal stereolithography (TSL)
	Non-metals	Melting and solidification	Heat	Heated nozzle	Fused deposition modelling Ballistic particle manufacturing
Solid	Metals	Melting and solidification	Heat	Electric Arc Laser beam Electrochemical discharge	Shape melting FDM FDM
	Thin sheets and foils	Selective gluing & cutting	Adhesive bonding	Glue & laser beam	Laminated object manuf.
	Semi polymerized plastic foils	Foil polymerization	Light	Lamp	Solid foil polymerization
Powders	Single component	Selective sintering	Heat	Laser beam	Selective laser sintering
	Coated powder	selective sintering	Heat	Laser beam	SLS
	One component and one binder	Selective powder binding	Chemical bond	Fine droplet beam of binder liquid	3D printing or MIT process powder selective binding

28/07/2024 Advantages

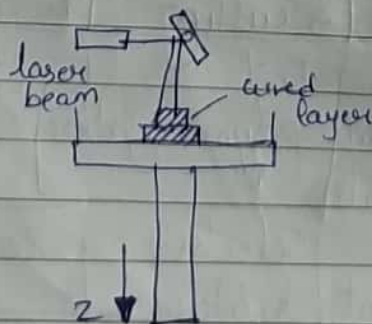
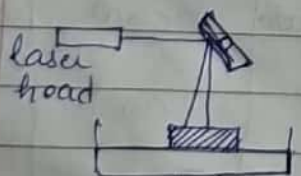
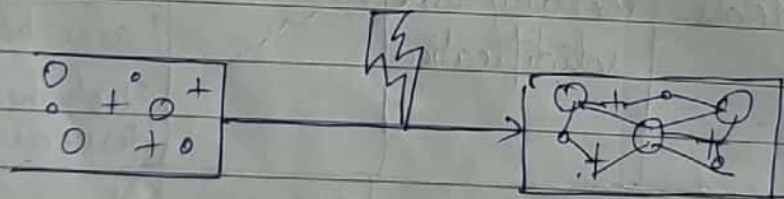
- 1) Design - basic dimensions reqd.
- 2) Planning - no need to define a blank geometry.
- 3) Tooling/Jigs/Fixtures - tool/less processes; no clamping required.
- 4) Automation -

29/8/24 Stereolithography

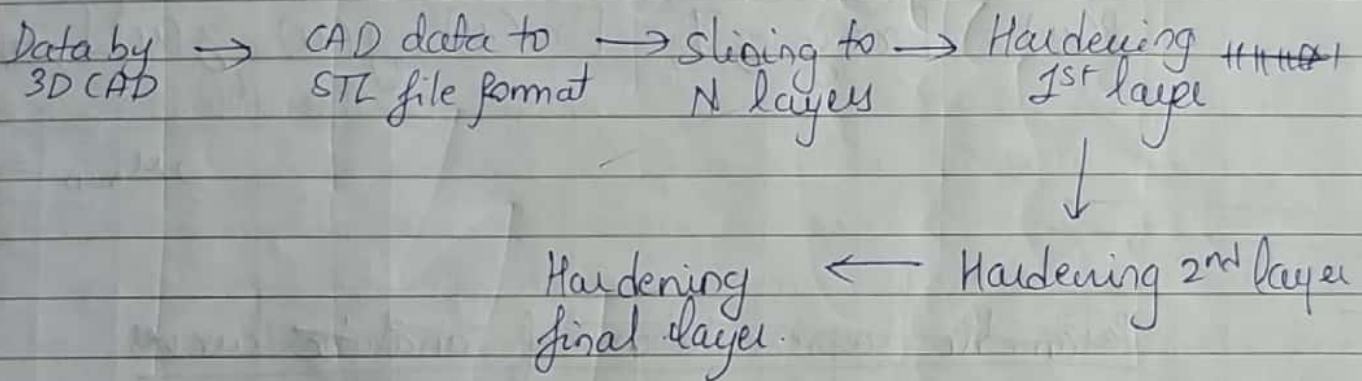
SLA

Principle.

- Part built from photocurable liquid resin. exposed to laser beam (photopolymerization) which scans across surface of resin.
- each layer is scanned by optical scanning system



- polymerization is process of linking small molecules (monomers) into chain like larger molecules (polymers)
- SLA is most commonly used.
- Good finish, good strength.
- layer by layer manufacturing.
- self adhesive property cause layers to bond.



30/8/24 Parametric curves
Curve representation

Explicit function - nonparametric

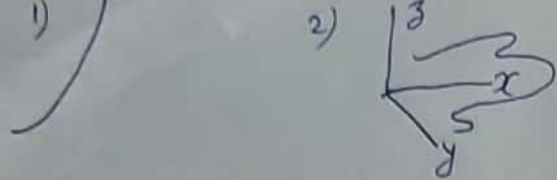
$$y = f(x) = 2x^2 + 1 \quad \left| \quad P = [x \ y \ z]^T = [x \ f(x) \ g(x)]^T$$

Implicit function - nonparametric.

$$y = f(x, y) = 2y^2 + 4xy + 3x + 4 \quad \left| \quad \begin{array}{l} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{array} \right.$$

Problems

- 1) slope of a curve at point is vertical or near vertical, its value becomes infinity or very large.



- 2) Shapes of most engg. objects are intrinsically independent of any coordinate system.
- 3) If curve is to be displayed as a series of points or segments straight line segments, the computations involved could be extensive.

parametric form

parameter 'u'

$$P(u) = [x \ y \ z]^T = [x(u) \ y(u) \ z(u)]^T_{u_{min}}$$

$$u_{min} \leq u \leq u_{max}$$

Parametric representation of analytic curves

June 2023

Suggest the additive manufacturing process for the casting company to make the patterns for investment casting made up from liquid thermoset polymer resins. Justify your answer with the working principle, advantages.

FDM4) Fill spacing

- It is the distance left between raster or contours.

* Process parametersSLA

- The laser beam diameter ~~if~~ is fixed at 200 μm on build surface for most of the SLA system.
- Because of scanning patterns, smallest feature that can be built around one & a half times the laser beam diameter (300 μm)
- In higher resolution systems such as Viper Si2 m/c, laser beam dia. is $70 \pm 15 \mu\text{m}$ and features as thin as 80 μm have been built using spot size.
- With smaller spot size capability adds small feature prevention.

2) Laser beam focus depth.

- In Z direction, cure depth of a layer is a fn of the focus depth underneath the resin surface.
- Thickness is inversely proportional ~~to~~ with cure depth, hence, depth of focus of laser beam
- Laser beam is focused beneath resin surface in order to increase Rayleigh range and accurate build platform.

size.

3) Layer Thickness

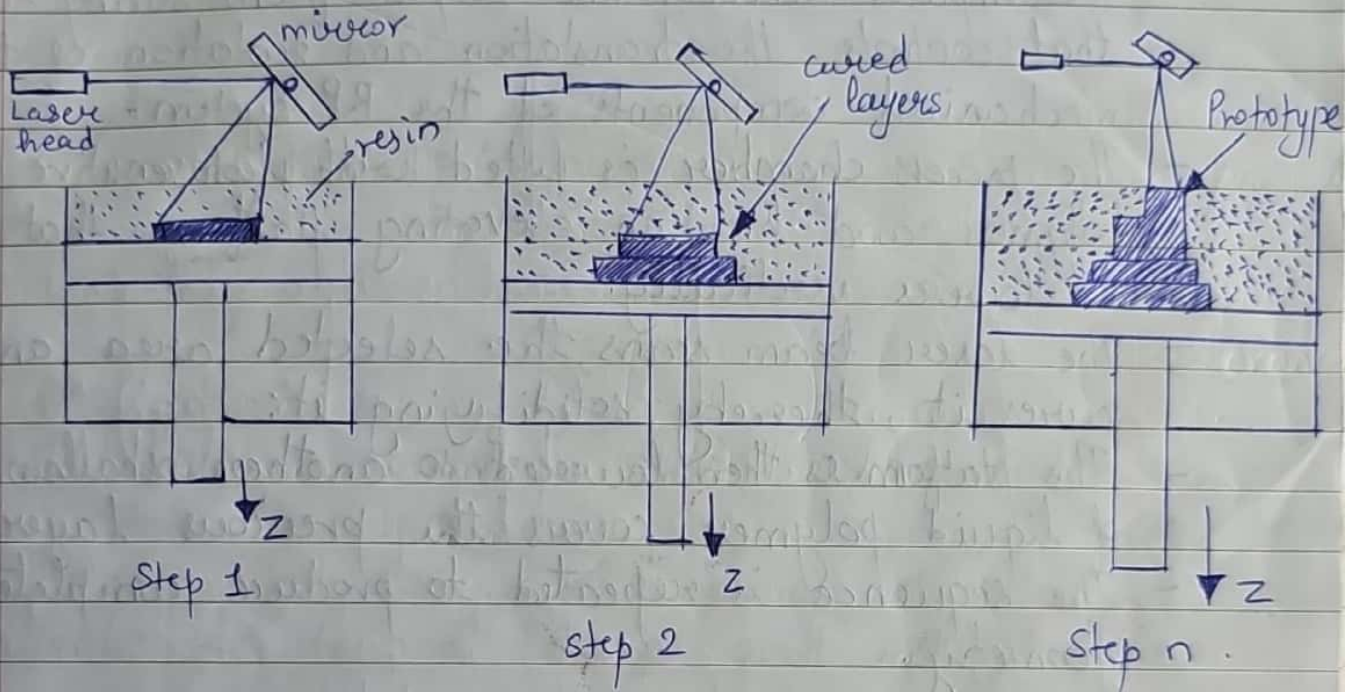
- Thickness of one layer is inversely proportional to depth of focus of the laser beam under the resin surface.
- Layer thickness is mainly limited by recoating mechanism and resin viscosity.
- This blade is necessary because SL resins have high viscosities, approx 180 - 2000 centipoise.
- Currently smallest layer thickness that can be obtained is 25 μm (0.001 inch).
- Layer thickness affects surface texture, accuracy in Z-axis and build speed.

1) Give the classification of Rapid Prototyping Process.

State of material	Type	Mechanism	Energy Type	Energy Source	Process
Liquid	Photo-polymers	Liquid photopolymerization	Monochromatic light	Lamp	Solid ground curing (SGC)
				Laser beam	Stereolithography (SLT)
				Holography	Holographic interference solid. (HIS)
			Light (two frequencies)	Two laser beams	Beam interference solid.
Solid	Thermo	Liquid Thermal	Heat	Laser beam	Thermal stereolithography
	Non metals	Melting & solidification	Heat	Heated nozzle	Fused deposition modelling (FDM)
	Metals	Melting & solidification	Heat	Electric Arc	Shape melting.
				Laser Beam	Fused deposition modelling
Powder	Thin sheets and foils	Selective glueing and cutting	Adhesive bonding and	Electrochemical discharge	Fused deposition modelling
	Semi-polymerized plastic foil	Foil polymerization	Light	Glue & laser beam	Laminated object manufacturing (LOM)
	Single component coated powder	Selective sintering	Heat	Lamp	Solid foil polymerization (SFP)
	one component and two powder	Selective sintering	Heat	Laser Beam	selective laser sintering (SLS)
			Chemical bonding	Laser Beam	selective laser sintering (SLS)
				Fine droplet beam of binder and selective powder	3D printing or MJT process

2) Explain with a neat sketch the principle of operation of stereolithography.

Ans



- Parts are built from a photo curable liquid resin that cures when exposed to a laser beam which scans across the surface of the resin.
- The building is done layer by layer, each layer being scanned by the optical scanning system and controlled by an elevation mechanism which lowers after the completion of each layer.
- Polymerization is the process of linking small molecules into chain like large molecules.
- When chainlike polymers are further linked to one another, a cross linked polymer is formed.

Working of SLA

- Data by 3D CAD → CAD data to STL → slicing to N layers
→ Hardening 1st layer → Hardening 2nd layer
→ Hardening final layer.

- Slicing software reads the CAD model and cuts it into thin layers that depend on the mechanical movement of the RP system and part geometry.
- The computer receives the output from slicing software that controls the translation and rotation of the mechanical components of the RP system.
- The process chamber is filled with photosensitive liquid polymer and has an elevating platform that raises or lowers vertically.
- The laser beam scans the selected area and cures it, thereby solidifying it.
- The platform is then lowered so another shallow layer of liquid polymer covers the previous layer.
- The sequence is repeated to produce complete part geometry.

Process parameters

- 1) Laser beam spot diameter - $200\ \mu\text{m}$
- 2) Laser beam focus depth (cure depth) - 8mm
- 3) Layer thickness - $25\ \mu\text{m}$
- 4) Viscosity of resin - $180 - 2000\ \text{cP}$.

Advantages

- 1) The SLA process can be used continuously & unattended round the clock.
- 2) The computerized support process serves as good user support.
- 3) SLA machines have wide range of build volumes.
- 4) It has good accuracy of $\pm 0.1\text{mm}$.
- 5) It has a good surface finish.
- 6) The process is of high resolution and can build complex

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- 6) The process is of high resolution and can build complex

parts.

Disadvantages

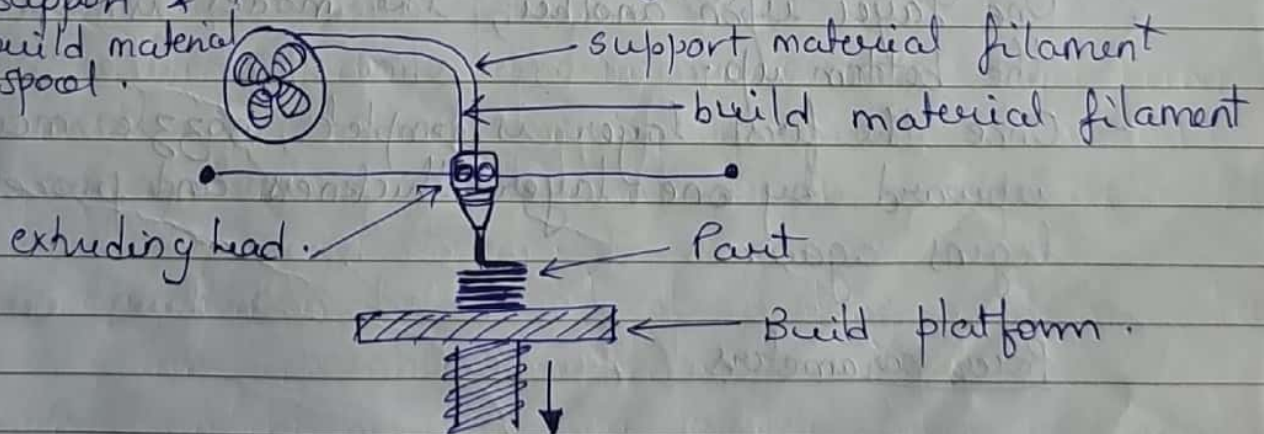
- 1) Structures that have overhang or undercut must be supported.
- 2) It requires post processing and cleaning.
- 3) Post curing is required to cure the object completely and ensure structural rigidity.

Applications

- 1) Patterns for investment casting, sand casting, molding.
- 2) Models for conceptualizing, packaging and presentation.
- 3) Prototypes for design, analysis, verification and testing.
- 4) Tools for jig and fixtures and tooling design.

Fused Deposition Modelling

support &
build material
spool.



Principle

- The ~~princ~~ FDM is based on surface chemistry, thermal energy and layer manufacturing technology.
- The material in filament form is melted in a specially designed head which extrudes the model.

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- As it is extruded, it is cooled and solidified.
 - The model is built layer by layer.

Working

- The process depends on building up the model layer by layer using a thermoplastic 0.0003 mm diameter filament.
- The filament is unwound from a coil and fed to the extrusion nozzle.
- The filament melts as it passes through the heated extrusion head and emerges as a thin ribbon.
- The FDM system OS imports the cad file.
- The system then orients the part for optimum building and slices the model.
- It creates a path generated by OS, laying down a very thin layer of molten material.
- One layer upon another, the model is built from the bottom up.
- Once the first layer is complete, nozzle moves upward by one layer thickness and process begins again.

Process parameters

- 1) Liquifier temperature = temperature is just above ~~heat~~ melting point of material.
- 2) Nozzle diameter = 0.012 mm to 0.025 in ch.
- 3) filament diameter = 0.0003 mm
- 4) Road width = diameter of the tip
- 5) fill spacing = 0 for solid part.

Advantages

- Quiet, nontoxic and office friendly process.
- Quick for small, thin parts.
- Produces strong parts.
- Post cleanup is not required.
- Produces multicoloured parts using ABS.

Disadvantages

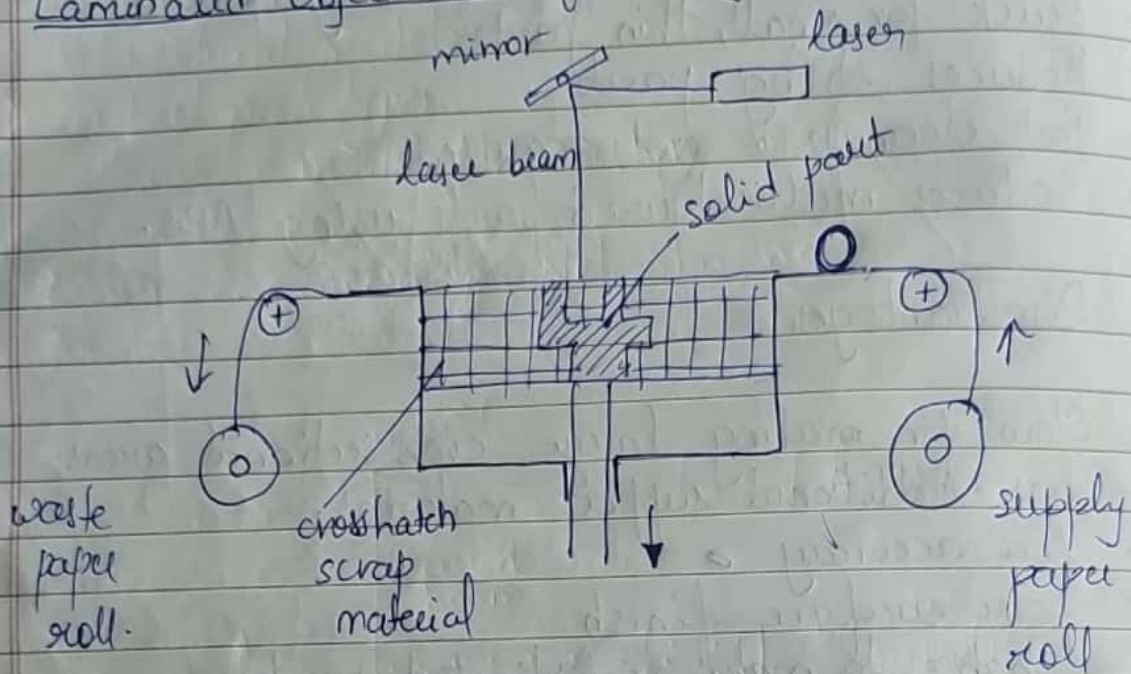
- Slow for making large cross sectional areas.
- has additional support material.
- low accuracy.
- poor surface finish.
- Unable to make complicated parts.
- Limited no. of materials available.

Application

- Models for conceptualization and presentation.
- Prototype for design, analysis, verification & testing.
- Patterns & masters for tooling.

Path Generation

- 1) CAD file preparation.
- 2) Part size.
- 3) Orientation.
- 4) Slicing.

Unit 2Laminated Object ManufacturingPrinciples

- Parts are built layer by layer, by laminating each layer of paper or other sheet like materials and contour of the part is cut by CO₂ laser.
- Each layer of the building process contain ~~one or more~~ cross sections of one or more parts.
- The next layer is laminated and built on top of the laser cut layer.

Working

- Once the parameters are loaded into LOM slice, the STL file slices as the part builds. This process of continuous slicing is known as ~~a~~ slice-on-the-fly.

- The paper is fed from a supply roller on the stack and first bonded to the previous layer using a heated roller which melts a plastic coating on the bottom side of the paper.
- The profile is then traced by an optic system mounted in the x-y plane.
- As the ~~page~~ laser has cut each profile, the roll of paper advances, a new layer is glued to the previous layer and the process is repeated.

Advantages

- 1) These include a wide variety of materials, such as paper, plastic, metal, composites, and ceramic.
- 2) The laser outlines the profile and not the entire surface area, thus build time is faster.
- 3) Support structure is not required as the part is supported by its own material.
- 4) Post curing is not required.

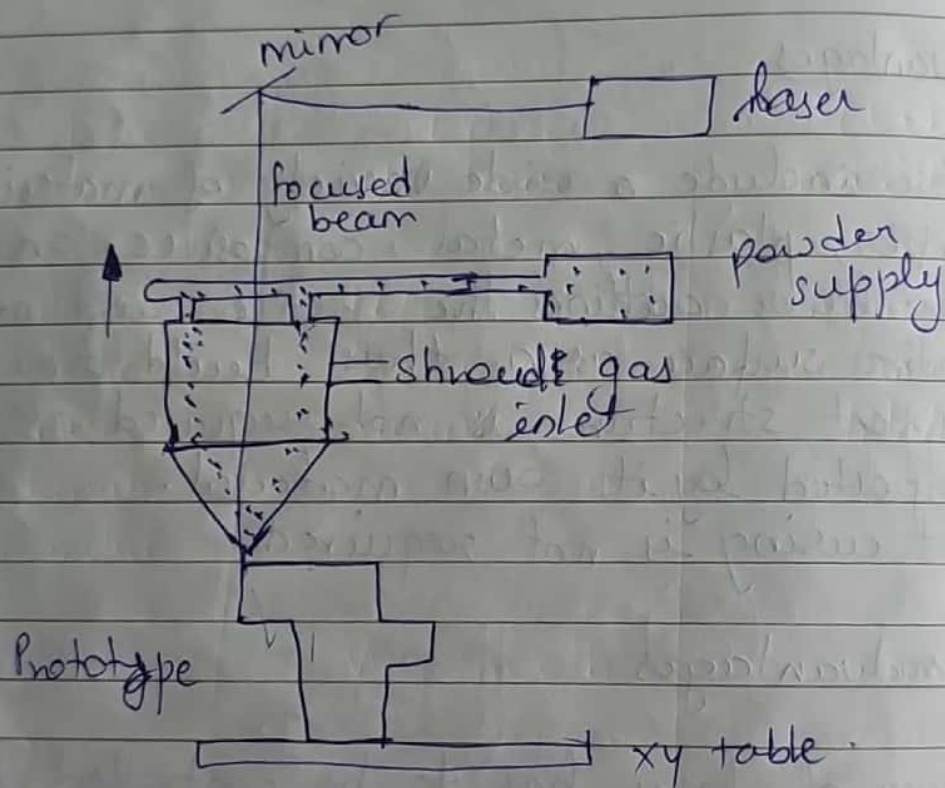
Disadvantages

- 1) Power of laser has to be controlled so that it does not penetrate previous layers.
- 2) Fabrication of thin walls in Z direction is not possible.
- 3) Strength of the part depends on the adhesive strength of the glue.
- 4) Removal of support material is tedious and time consuming.

Applications

- 1) Visualization (Packaging & Design)
- 2) Architectural models
- 3) Product prototypes
- 4) Patterns & molds for casting

Laser Engineering Net Shaping



Principle

- A high powered nd:YAG laser focused on a metal substrate creates a molten puddle on the substrate surface.
- Powder is then injected into the molten puddle to increase material volume.
- A printing motion system moves the build platform horizontally & vertically as the laser beam traces

the profile.

Process

- STL file is sliced and downloaded to the machine from bottom slice up.
- A deposition head supply metal powder to the indirect laser beam to be melted.
- Laser is directed by optical mirror.
- Fabrication process takes place in low pressure argon chamber to minimise for oxygen free operation.
- When a layer is complete, the deposition head moves up and continues with the next layer.
- The process is repeated layer by layer until the part is completed.

Materials

- Inconel, copper, aluminium, titanium.

Advantages

- It produces fully dense metal parts with good grain structure.
- Produces complex metal parts.
- Post processing is minimized, reducing process time.

Disadvantages

- It produces only metal parts.
- Large physical unit size.
- Requires high power consumption.

Selective Laser Sintering (SLS)

Materials

- Polyamide
- Thermoplastic elastomer
- Polycarbonate
- Nylon
- Metal

	SLS	SLA	LENS
Laser Used	CO ₂ laser	UV laser	nd:YAG laser
Material	Polyamide, thermoplastic elastomers, Polycarbonate, nylon, metal	PLA, Photopolymer resins	Inconel, copper, aluminium, titanium
Surface finish	Rough	Very smooth	Fair
Support structure	Not reqd	required	not reqd

	Solid based	Powder based.
Material	solid filaments sheets or blocks	Fine powder
Technique	LOM, FDM	SLS
material handling	fed continuously	sprayed in layers.
wastage	minimum	high
support	required	may or may not required.
Cost	low cost	high cost
Post processing	required	not reqd.

	RP	Conventional.
Method	Additive	Subtractive
Lead time	Short	Long
Material usage	Efficient	more waste
Volume	Low volume production	High volume production
Cost	low initial cost	High initial cost

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Rapid Manufacturing Process Optimization.

Factors influencing Accuracy

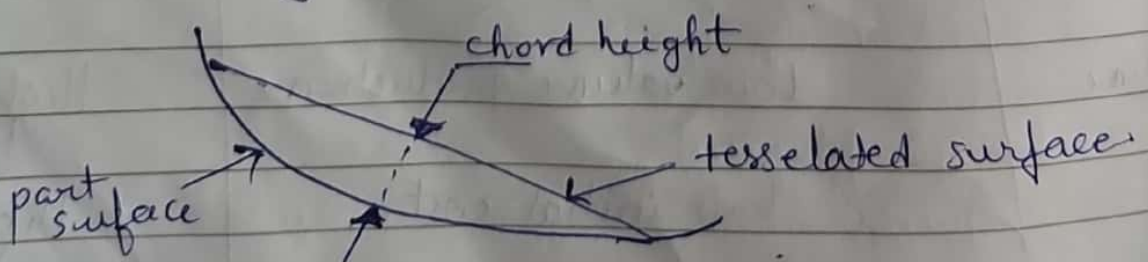
- Factors are independent.

- 1) Factors causing errors during the data preparation such as STL file generation, model slicing and part build direction
- 2) Factors influencing the part accuracy during the build stage such as specific parameters.
- 3) Factors directly related to the part finishing techniques such as post processing steps.

1) Data preparation Error (Error due to tessellation)

- STL file approximates surface of the 3D model by triangles.
- Tessellation error can be minimized by increasing no. of triangles.
- no. of Δ 's cannot be increased indefinitely
- STL generation process can be controlled by specifying

- i) Chord height.
- ii) Angle control factor.



- less deviation, smoother curve, small chord height.

- Angle control specifies required definition level along curves with small radius.

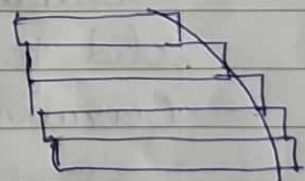
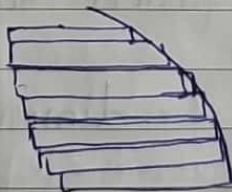
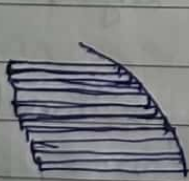

smaller radius
should be tessellated


Bigger radius should not
be tessellated.

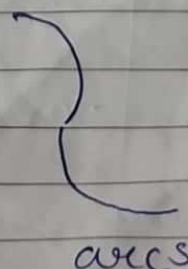
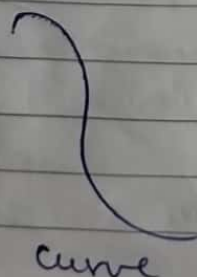
- It defines a threshold for curve radius (r_0) below which curves should be tessellated.

2) Error Due to Slicing

- Stair stepping is the consequence of addition of material in layers.

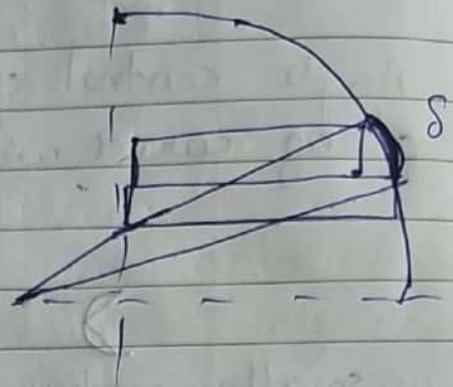


- curves can be replaced with arcs.



- error due to replacement of a circular arc with stair steps is

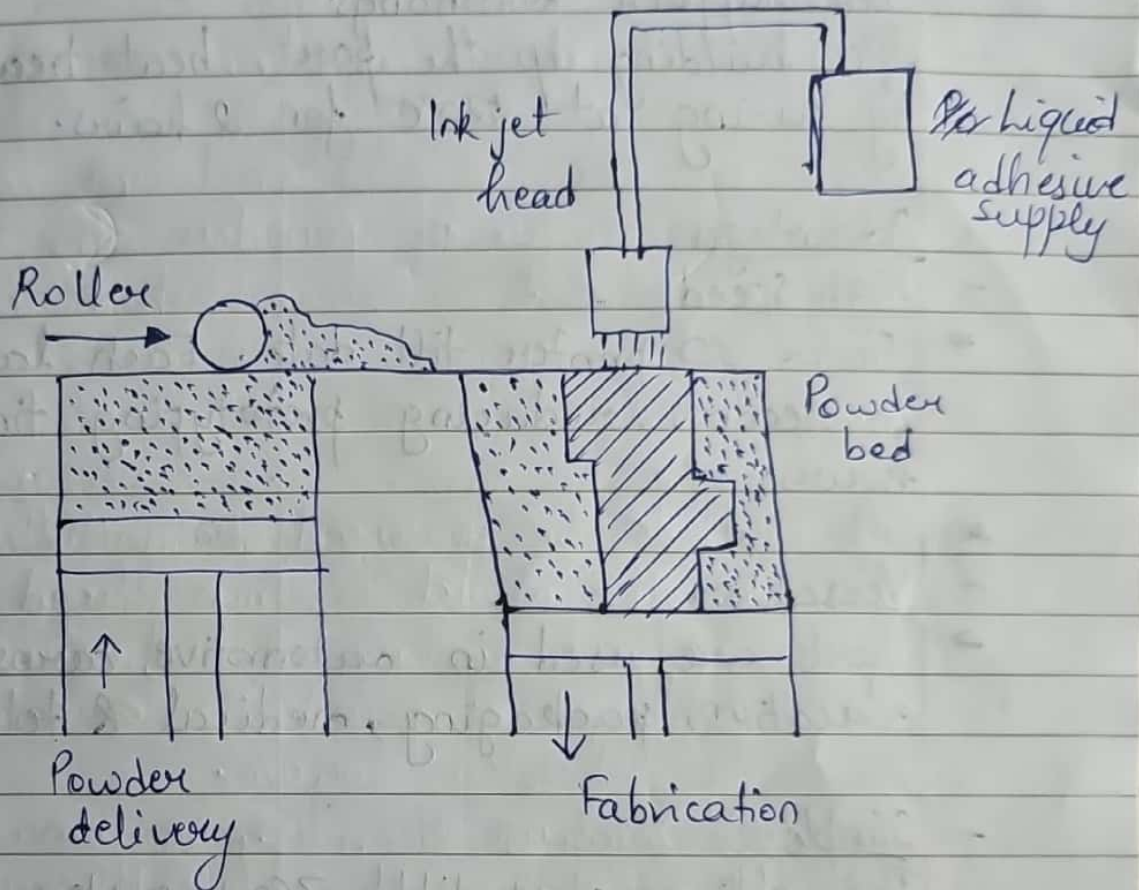
$$\delta = r - l$$



Part Building Error

- 1) Curing Error
 - Errors caused by over curing & scanned line shape.
- 2) Control Error
 - Errors caused by layer thickness & scan position control.
 - Overcuring is required to adhere layers to form solid parts.
 - Overcuring causes dimensional & positional errors.

Scanned line shape

Unit 2I) 3D printing

- The process starts by depositing a layer of powder object material at the top of the fabrication chamber.
- A measured quantity is supplied from a similar supply chamber by a piston moving upwards.
- The roller distributes and compresses the powder at the top of the fabrication chamber.
- The inkjet head deposits a liquid adhesive in a 2 dimensional pattern on the layer of powder.
- It bonds in the ~~top~~ area where adhesive is deposited to form a layer of the object.
- Once a layer is complete, the fabrication piston moves down by one layer thickness and the process is repeated until entire object is formed.

- After completion, the object is elevated and extra powder is brushed off.
- No external supports are required because powder bed supports overhangs.
- After building up the part, heat treatment is applied by curing at 120°C for 2 hours.

Advantages

- High Speed
 - Faster 3D printer till date. Each layer is printed in seconds, reducing prototyping time to 1 to 2 hours.
- Versatile
 - Parts are used in automotive, aerospace, footwear, education, packaging, medical & telecom industries.
- Simple

The office compatible ZCorp system is easy to operate and does not require a designated technician to build a part.
- No wastage of material.

Powder which is not printed during the cycle can be reused.
- Color

Enables color schemes from a 24 bit palette of colors.

Disadvantages

- 1) Limited functional parts
Relative to SLS, parts built are weak and limits functional testing capabilities.
- 2) Limited materials
Material available are only starch and plaster based materials, with the added option ~~of~~ to infiltrate wax using ZW4 Waxer.
- 3) Poor surface finish
Parts built by 3D printing have relatively poorer surface finish.

Applications

- 1) Adidas has used this to reduce prototype development time and communicate in new ways.
- 2) Parts produced find application ~~in~~ with computer game creators, animators, pre hollywood mockup designers and sculpturing artists.

Electron Beam Melting

Principle

- Parts are built up when an electron beam is fired at the metal powder.
- Computer controlled electron beam in vacuum melts the layer of powder as per CAD model.
- Building of part is accomplished layer by layer.

Process

- The part to be produced is modelled in 3D CAD program. The model is then sliced into thin layers, approx a tenth of millimeter thick.
- A thin layer of powder is scraped on the vertically adjusted surface.
- The first layer's geometry is created from the layer of powder melting at the points specified by the CAD file with the electron beam.
- The building surface is lowered & next layer of powder is placed on top of the previous layer.
- This procedure continues repeats until a finished metal part is completed.

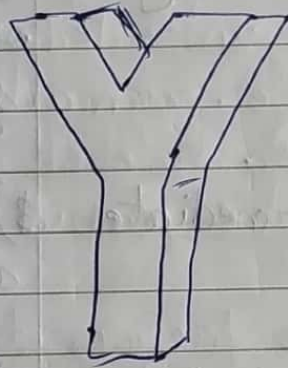
Components:

- 1) Electron beam gun with sweeping system.
- 2) Vacuum tank with powder holder and fabrication tank
- 3) Vacuum pump
- 4) Monitor
- 5) Linear device

- 6) High voltage unit
- 7) Electronic control system
- 8) Control unit.

QB IT 2 (OE)

- 1) Diff. b/w direct & indirect tooling.
- 2) Explain concept of part orientation & support generation for foll. 3D part.



- 3) Explain part building errors in SLS & how to compensate for it?
- 4) Explain importance of proper part build orientation
- 5) Diff. b/w silicon rubber tooling & aluminium

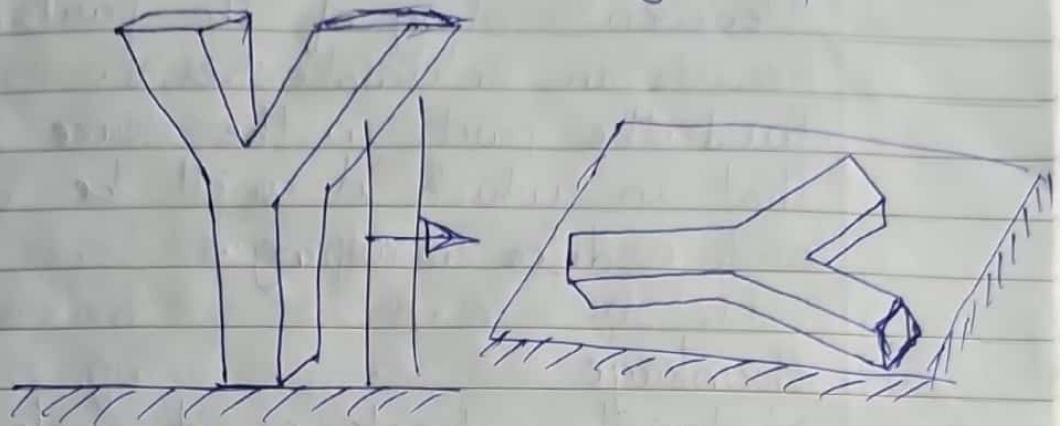
IT-2 Question Bank

① Differentiate between Direct & Indirect Tooling

Ans

Direct Tooling	Indirect Tooling
1) Final tool/mold is created directly from the fabricated part.	1) Pattern/model is created from the fabricated part, producing the final tool.
2) Involves CNC machining or 3D printing.	2) Involves casting, molding or using patterns for shaping.
3) Faster, skips intermediate steps.	3) Slower, requires additional steps.
4) High precision due to direct creation from the design.	4) Precision depends on the quality of pattern and process.
5) Less durable for long time use.	5) Best for More durable for high volume product
6) Suitable for low volume production.	6) Best for high volume production due to high durability.
Ex. 3D printing a mold for injection molding.	Ex. Creating a wax pattern for investment casting.

- ② Explain the concept of part orientation and support generation for the following 3D part



Ans Minimize the distance between the part and CAD origin.

2

- Ans - The 3D part is vertical, making the distance between the part and the CAD origin high.
- Part orientation is in XZ plane.
 - Rotate the part in the XY plane and making it flat.
 - This will reduce the overall height of the object.
 - This will reduce the numbers of layers to be built.
 - Support generation will not be required in the XY orientation as there are overhanging parts.

- 3) Explain building part errors in SLS and how to compensate for it?

Ans The orientation constraints of the SLS process depend on

- i) The process accuracy in X-Y plane and Z direction
- ii) Materials used, since shrinkage and anisotropic properties of the sintered powders are different.

- Most common building errors are:

i) Warping

- Uneven cooling of the part during the process results in internal stresses.
- Build the part in the same orientation as that in which it will be cured. This will reduce warping.

ii) Shrinkage

- Material contracts as it cools, and if not compensated for, the parts dimensions will be smaller than intended.
- Parts should be normally angled by 10° - 15° around X-axis. Rate of change of cross sectional area is not too rapid to ensure consistent shrinkage.
- Apply a scale factor to the design to account for expected shrinkage.

iii) To maximize strength, parts are to be built in the XY plane to improve flexibility.

iv) Critical features should be built on upward facing model surfaces to have best feature definition.

v) Overhangs should be minimized when building with polycarbonate, sand and elastomers.

~~vi)~~

④ Explain the importance of proper part building orientation.

Ans - The build should be oriented to :

- 1) Minimize height of support structure.
- 2) Minimize distance between part and CAD origin.
- 3) Minimize overall height of the object.
- 4) Minimize the number of slanted or sloping surfaces.

- If a surface is oriented to require fewer support structures, it can have a smoother surface and accuracy.
- Orienting and aligning the layers properly ensure better strength and durability.
- By selecting right orientation, one can minimize the use of support material and reduce material usage.
- Proper orientation helps manage heat distribution and internal stresses.

⑤ Differentiate between silicon rubber tooling and aluminium epoxy tooling in detail.

Ans

Soft

hard

Silicon Rubber Tooling

Aluminium Epoxy Tooling.

- | | |
|--|--|
| 1) Made from silicon, a flexible elastomer. | 1) Combination of aluminium and epoxy resin. |
| 2) Highly flexible, can easily stretch and bend. | 2) Rigid and inflexible, provides structural strength. |
| 3) Easier to remove complex parts due to its flexibility | 3) More difficult to release parts due to rigidity. |

4) Moderate thermal resistance	4) High thermal resistance
5) Provides smooth surface finish.	5) Provides a more precise surface finish.
6) Lower cost for small production runs.	6) Higher initial cost due to materials and manufacturing efforts.
7) Faster to produce, used for quick prototyping.	7) Takes longer to produce due to precision machining.

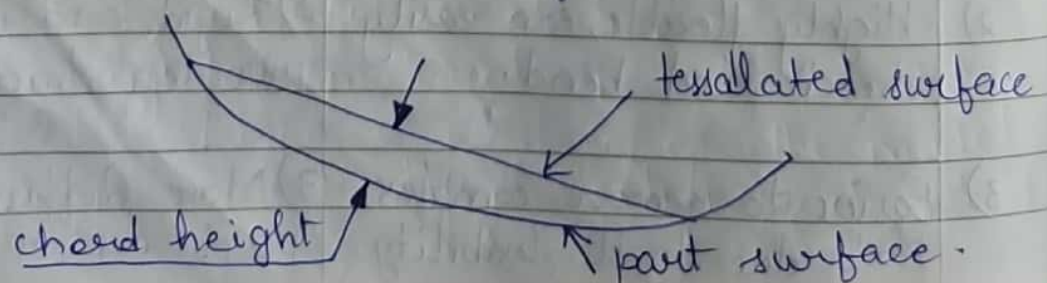
6) Write a note on factors which influence the accuracy of the additive manufactured component.

Ans 1) Data preparation errors

- Errors due to tessellation.
- A STL file approximates the surface of 3D CAD model by triangles.
- Tessellation errors can be minimized by increasing the no. of triangles.
- STL generation process can be controlled by specifying

i) Chord height

- It is the maximum distance between part surface and tessellated surface.



- Decrease of chord height leads to smoother surfaces, but increases file size.

ii) Angle Control

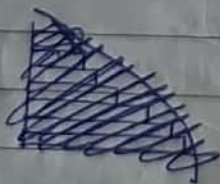
- It defines a threshold for the curve radius below which curves should be tessellated.



Small radius' should be tessellated

Bigger radius' should not be tessellated.

- Errors due to slicing
 - Stair stepping is a consequence of addition of materials in layers.
 - Curves are described with curvatures and curvature radius.
 - Curves of a CAD model can be replaced by arcs to assess the error of stair steps



2) Part Building Errors

a) Curing Errors

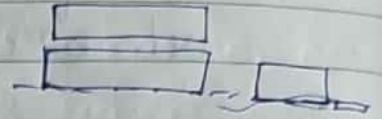
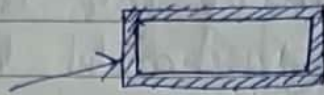
i) Overcuring

- Laser over curing is required to adhere layers

to form solid parts.

- Overcuring causes dimensional and positional errors.

Error basically curing outside of what is needed.



ii) Scanned line shape

- Scanned line is created when laser beam scans surface of resin.
- The crosssection of scanned line is referred to as scan line shape.

b) Control Errors

- Layer thickness should be ^{at the} defined value and the border line ~~so~~ should be positioned at specified positions.
- The layer thickness is variable and border position is not precise.

3) Finishing Errors

a) Varying amount of material.

- Each RP process produces corners as the stair steps with different resolution.
- The amount of material that has to be removed to improve surface of the model can vary due to the selected part build orientation.

b) Finishing technique

- Techniques with better dimensional control will have less impact on model accuracy.
- Models finished employing milling will have less influence on accuracy than those using manual wet sanding or sand blasting.

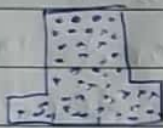
- 7) Write a short note on sand casting tooling.
- Ans - RP models can be used as patterns for sandcasting.
- A sand pattern consists of sand mixed with bonding agents, which is all contained in a flask.
 - A model is used to make impression in two halves or the cope & drag of sand pattern.
 - Molten metal is cast into this sand impression, allowed to cool, then pulled from cavity & process is repeated.

Steps

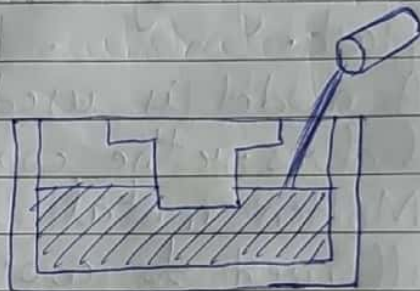
- a) Inverted drag box is placed on the floor.
- b) Pattern is placed in the box.
- c) Then sand is poured into the box.
- d) Sand is rammed and levelled off.
- e) Then the drag box is turned over.
- f) Cope box is placed over it.
- g) Second half of pattern is placed.
- h) Sand is poured in the box.
- i) Sand is rammed and levelled off.
- j) Now two boxes are opened.
- k) Then pattern is removed.

8) Explain 3D Keltool process in detail.

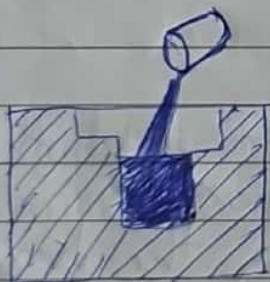
- Ans - This process is based on metal sintering process.
- The process starts with a RP pattern from which a rubber mould is created.
 - The rubber mould is then used to cast into the mould geometry which after hardening is in the green state.
 - The green mould is fired and copper infiltrates resulting with a tool (70% steel & 30% copper).
 - Ejector pins, cooling lines & other accessories may be added in final machining and polishing steps.



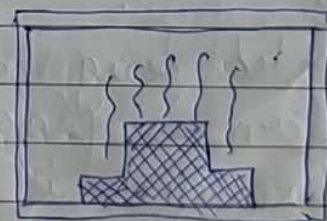
I) Master pattern



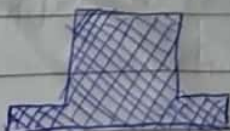
II) Silicon casting



III) Casting with tool steel/Tungsten carbide/epoxy mixture.



IV) Burn-out of binder, sintering and infiltration with copper in an oven.



V) Tool insert ready for production.

(10) Write a note on direct AIM method?
Ans - AIM - Accurate clear epoxy solid Injection Moulding.

- Mould is grown on a SLA system, using the ACES build style, that is shelled out on the bottom side.
- This leaves a cavity in the mold halves that can be backfilled with aluminium filled epoxy, ceramics, and low melting metals.
- The backfilling process provides a thermal conduit for the heat exchange system, as well as integrates the cooling system that may be put into the mold halves.
- These mold halves are mated and aligned, and the part surfaces are finished for surface quality.
- Using extended cycle times and a release agent, numerous parts can be made directly by injecting the final thermoplastic into the ACES mold core and cavity halves using a standard injection molding machine.

(11) Write a note on copper polyamide tooling.

- Ans - Copper polyamide (copper and plastic) is a new metal plastic composite designed for short run tooling applications.
- Tooling inserts are directly produced in the SLS machine with a layer thickness of 75 μ m.
 - The insert surface is sealed with epoxy and finished with sandpaper and finally shell inserts are packed up with a metal alloy.

15) Explain the solid ground curing (SGC) process.

The process has two cycle:

- Photomask generation cycle.
- Layer fabrication cycle.

Principle

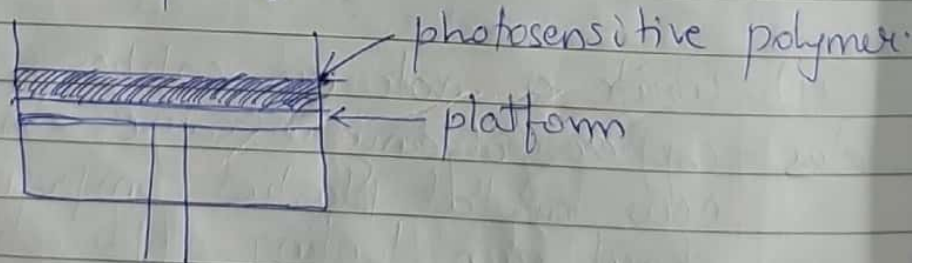
- Parts are built layer by layer, from a liquid photopolymer resin that solidifies when exposed to UV light.
- The irradiation source is a high powered collimated UV lamp and the image of layer is generated by masked illumination.
- The mask is created from CAD data input and printed on a transparent substrate.

Steps

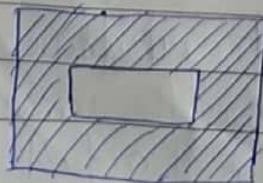
- CAD file is sliced



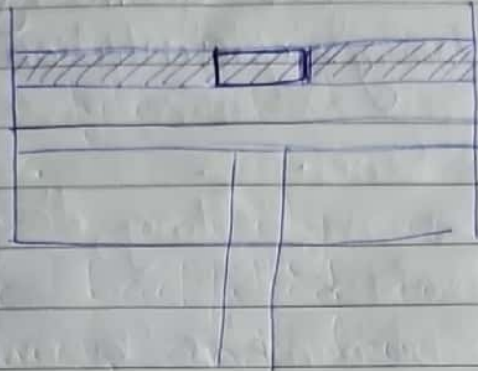
- A layer of photosensitive polymer is first sprayed onto the platform.



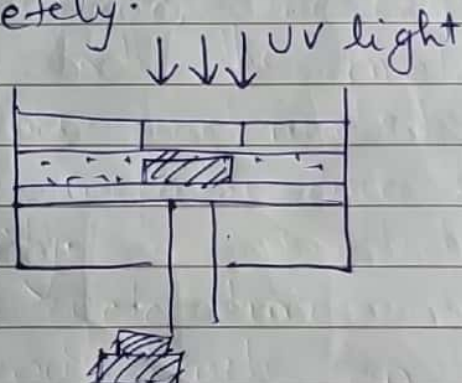
- Photomask is prepared for each layer using electrophotography.



- 4) The mask plate is placed on top of platform and object is under construction.



- 5) UV light is ~~expose~~ exposed onto the layer, which solidifies completely.



- 6) The residual polymer is vacuummed for recycling and replaced by wax.

- 7) The wax is hardened using the wax cooling plate.

Unit 4

1) Computer Aided Tissue Engineering (CATE)

- Tissue engg. is used to replace failing or malfunctioning organs such as skin, liver, pancreas, ligaments, cartilages or bones.
- CATE advances modelling, design and fabrication of tissue scaffolds.
- It can apply biomimetic (biomimicry) design approach to introduce multiple biological and biophysical requirements in the scaffold.
- It also integrates biomimetic and nonbiomimetic features into the scaffold modelling database to form high-fi and smart scaffolds.
- It is based upon real anatomical data scanned by MRI/CT images or can be purely created in a CAD environment.

2) What is the role of AM in customized implants & prosthetics?

- Ans- Computer Aided Tissue Engineering is used to replace failing or malfunction organs such as skin, liver, pancreas, ligaments, cartilages and bones.
- CATE advances modelling, design and fabrication of tissue scaffolds.
 - Rapid Prototyping has been used to create spacers for spinal fixation problems by fabricating 3D structures with good interconnecting pores for tissue growth.
 - By using RP and CT scan, a 3D CT data file is transferred to RP system for building the

mold for a cranium implant.

3) Write a short note on STL file errors.

Ans - Standard Tessellation language contains no topological data.

- Tessellation algorithms are used to create approximate models which exhibit the foll. types of errors:

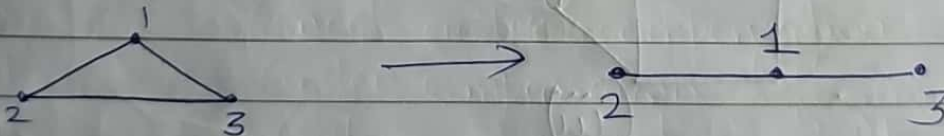
i) Missing facets or gaps

- Tessellation of surfaces with large curvature can result in errors at the intersection between the tessellated surface and the ^{part} original surface, leaving gaps or holes.

ii) Degenerated facets

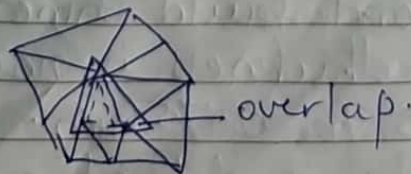
~~o all the~~

- It occurs when all of the facet's edges are collinear even though all its vertices are ~~distinct~~ distinct.



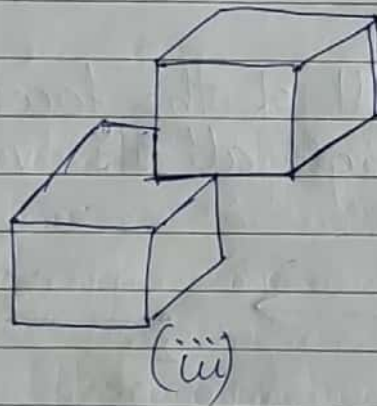
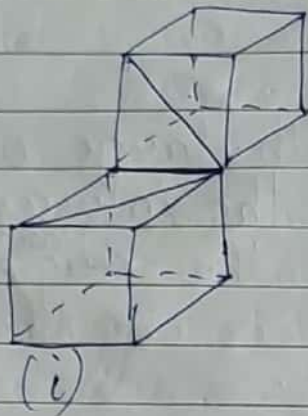
iii) Overlapping facets

- They are generated due to numerical roundoff errors during tessellation.



iv) Non-manifold topology conditions.

- They are generated because tessellation of fine features are susceptible to round off errors.
- There are three types:
 - i) A non-manifold edge.
 - ii) A non-manifold point.
 - iii) A non-manifold face.



4) What is model slicing?

Ans - The slicing program converts the 3D object in STL file into 2D cross sections.

- The thickness of the slice defines the texture, accuracy and build time.
- The layer thickness is normally in the range of 0.0625 to 0.75 mm.
- The slice axis is defined as the normal to the plane created by slicing.

5) What is point cloud? Write a note on generation of cloud using point cloud.

Ans - A point cloud is a set of data points in some coordinate system.

- In 3D coordinate system, the points are defined by the X, Y, and Z coordinates.
- Point clouds are created by 3D scanners.
- These devices measure a large no. of points on an object's surface and gives a point cloud as a data file.
- Point clouds are used to create 3D CAD models for manufactured parts, quality inspection and for visualization, rendering, animation and customization.

6) Collaboration Tools

a

6) Data Processing for Additive manufacturing technology.

i) Part Orientation

- Orientation influences build time, part resolution and surface finish.
- Higher resolutions can be obtained in curved surfaces by orienting them in horizontal plane.
- 'Stair-step' problem can be avoided by proper choice of orientation.

ii) Support generation

- While slicing the model into layers, isolated islands are formed which belongs to a projection from

- the main body.
- The connection of the isolated island will be built later.
- Therefore it is important to provide support to the isolated island to prevent it from falling due to gravity.
- Supports are also essential for cantilevered projections.

iii) Model slicing

- refer to Q. 4).

iv) Surface digitization.

- Technology used for measurement of surface of object are:
 - i) System based on laser scanning
 - ii) System based on white light projection.
- Equipment used is different but principle is same: triangulation.
- The laser, light sensor and object surface form a triangle.
- When the difference between laser and light sensor are known, distance of the object surface to the scanning system is known.
- White light projectors employ projectors instead of lasers.
- A triangle is formed between projector, camera & object surface.

v) Surface generation from point cloud

- refer to Q.5).

vi) Surface modification

- data is ~~transferred~~ to transferred to solid models.
- Transfer of data is necessary as one organization can develop the CAD model, while another can perform analysis on the same model, and a third manufactures the product.

7/i) Collaboration tools

- Collaboration tools streamline the design, prototyping and production processes by facilitating the sharing of 3D models, modifications and feedback among team members.
- Key tools include GrabCAD and Fusion360 which allow designers to work on the same file simultaneously.
- ANSYS additive suite help stimulate & optimize designs before printing.

ii) Internet based RP softwares

1) Mimics-Z

- Doctors & technicians can create 3D models from MRI or CAT scans easily.
- ~~Before~~ It is optimized for printing full colour 3D models in few hours.

- It allows ~~you~~ to process & edit ~~your~~ 2D image data to construct 3D models.
- It allows to take measurements and segment CT/MRI images.
- It also facilitates FEA, design, ^{surgical} simulation and additive manufacturing.

2) Solid View

- It allows non-CAD users to view, measure, translate and markup CAD data.
- It opens communications to all who need to be involved in design process.
- It is a low cost solution to access CAD data to engineers, scientists, technicians, designers, managers.

Advantages

- 1) Virtual review of complex part & assemblies.
- 2) Design collaboration across long distances, time zone and CAD systems.
- 3) Access to CAD data for technical illustration, packaging and marketing.

3) MAGICS

- It is a rapid prototyping software and key element of Magic's e-solution suite.
- It streamlines, automates and boost the rapid prototyping & manufacturing process.
- It enables to import a wide variety of CAD formats (.DWG, .DXF, .STL, .3ds, .DGN)

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- It exports STL file ready for rapid prototyping & manufacture.

Applications

- i) repairing & optimizing 3D models.
- ii) analysing parts.
- iii) documenting projects.
- iv) production planning.

4) Magics Communicator

- It views STL, IGES, DXF files with fast rotation, zooming & cross sectioning.
- It adds 2D & 3D annotations, text, shapes.
- Easily create 2D drawings to 3D files.
- Allows measuring of distances, radii & angles.
- Makes 3D slideshows for presentation.
- Ease of use as non CAD users can also be comfortable with the software.

Unit 1

1) Classification of RP.

State of material	Type	Mechanism	Energy Type	Energy Source	Process.
Liquid	Photo polymer	Liquid photopolymerisation	Monochromatic light	Lamp	SGC
			Two Lights	Laser	STL
				Holography	Holographic interference solid BLS
Thermo polymer	Liquid thermal	Heat	Heat	Two laser beam	Thermal STL
				Heated nozzle	FDM
				Ballistic particle manufacturing	
Metals	Melting & solidification	Heat	Heat	Electric arc	shape melting
				Laser beam	FDM
				Electrochemical discharge	FDM
Solid	Thin sheets	selective glueing & cutting	Adhesive bonding & cutting	Glue & laser beam	LOM
	Single component coated powder	Selective sintering	heat	laser beam	SLS
				laser beam	SLS
One component & one binder	Powder binding	chemical bond	chemical bond	fine binder liquid	3D printing

2) AM vs. conventional processes

1) Design

- Other manuf. processes require detailed analysis of part geometry to determine the step of fabrication, tool ~~reqs~~ and process requirement.
- AM only needs basic dimensional data & small knowledge of how AM system works.

2) Planning

- There is no need to define a blank geometry as no blanks are required.
- Minimum operations or process planning. GMP's are based on one operation only.
- No complicated scheduling or routing.

3) Tooling/Jig-Fixtures

- GMP's are toolless processes.
- No clamping devices, jigs & fixtures required.
- Mold dies & tool design eliminated.

4) Automation

- Shape generation process is generalized in character.
- It is done by using a computer.

3) Role of 3D softwares in AM.

1) Design creation

- Allows to create a digital object model of the object to be manufactured.

ii) Customization

- Allows for rapid customization to meet specific design requirements.

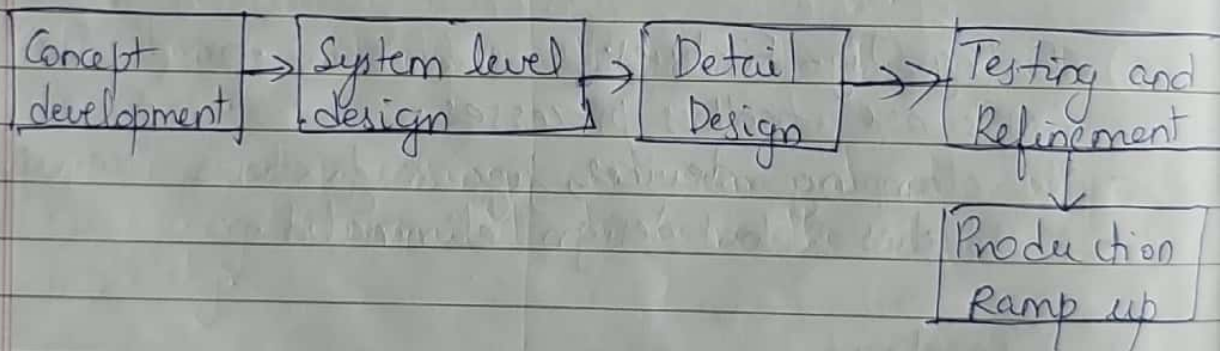
iii) Simulation & testing

- Simulation feature allows to analyze strength, durability & performance of design before printing.

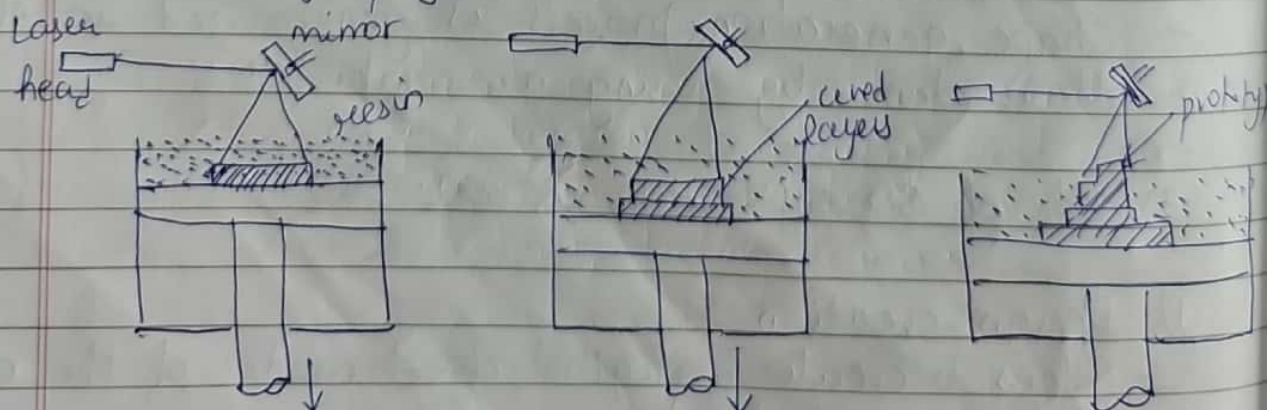
iv) File export for printing

- Once design is complete, it is exported to file format compatible with 3D printers and slicing softwares.

4) AM for new product development



5) Stereolithography (STL)



- Parts are built from a photo curable liquid resin which cures when exposed to a laser beam which scans across the surface of the resin.
- The building is done layer by layer, each layer being scanned by optical scanning system, and controlled by an elevating mechanism which lowers after completion of a layer.
- Polymerization is the process of linking small molecules into chain like larger molecules.
- When chain like polymers are further linked to one another, a crosslink polymer is formed.

Working of STL

- Data by 3D CAD → CAD data to STL → slicing to N layers → Hardening 1st layer → Hardening 2nd layer → Hardening final layer.
- Slicing software reads the CAD model and slices it into thin layers depending on the mechanical movement of the RP system and part geometry.
- The computer receives the output from slicing software and controls the translation & rotation of mechanical components of the RP system.
- The process chamber is filled with photosensitive liquid resin and has an elevating platform which raises or lowers vertically.
- The laser beam scans the selected area and cures it, thereby solidifying it.
- The platform is lowered so that another shallow layer of resin covers the previous layer.
- Sequence is repeated to complete the part.

Process parameters

Laser beam dia = 200 μ m

depth of focus = 8mm

Layer thickness = 25 μ m

Resin viscosity = 200-2000 cP

Advantages

- 1) SLA can be used continuously & attended round the clock.
- 2) Computerized process provides good user support.
- 3) It has a good accuracy of ± 0.1 mm.
- 4) It has a good surface finish.
- 5) Process is high resolution & builds complex part.

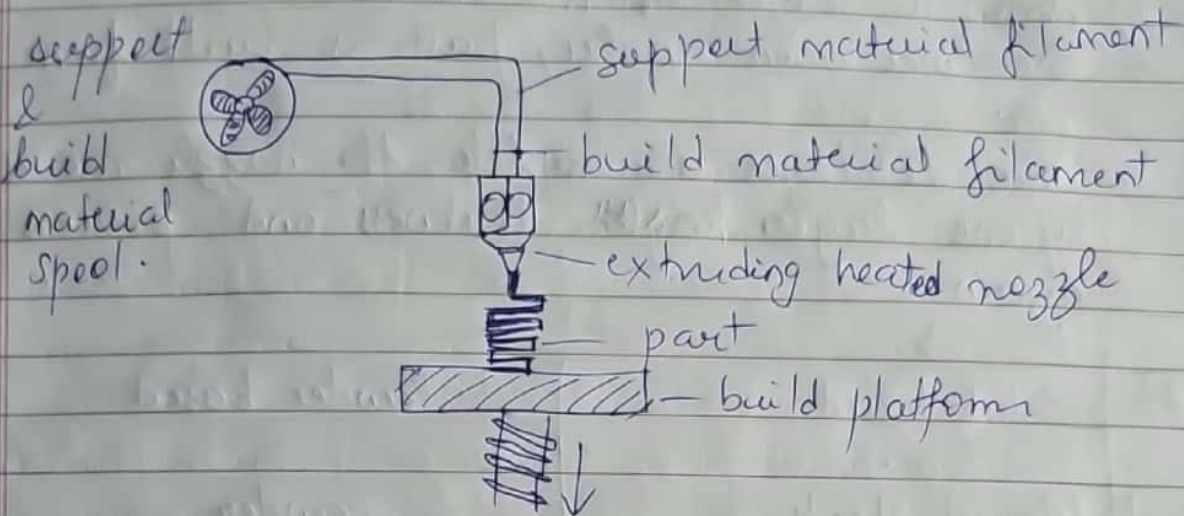
Disadvantages

- 1) Overhangs and undercuts have to be provided supports.
- 2) Requires post processing and removal of supports.
- 3) Post curing is required to cure the object completely.

Application

- 1) Patterns for investment casting, moulding, sand casting.
- 2) Models for conceptualization, packaging and presentation.
- 3) Prototypes for design, analysis, verification & testing.
- 4) Tools for jigs & fixtures, tooling design.

6) Fused Deposition Modelling (FDM)



Principle

- FDM is based on surface chemistry, thermal energy and layer manufacturing technology.
- The material in filament form is melted in a specially designed head which extrudes on the model.
- As it is extruded, it is cooled and solidified.
- Part is built layer by layer.

Working

- The process depends on building up the part layer by layer using a thermoplastic 0.0003 mm dia. filament.
- The filament is unwound from a coil & fed into the extrusion ~~model~~ nozzle.
- The filament melt as it passes through the heated nozzle and emerges as a thin ribbon.
- The FDM system imports the CAD file
- The system orients the part for optimum building & slices the part into n layers.

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- It creates a path generated by the system and lays down a layer of molten material.
 - One layer upon another, part is built from bottom up.
 - Once the layer is complete, the nozzle moves up by one layer height thickness and process starts again.

7)	Solid based	Powder based
Material	filament, sheets	powder
Technique	LOM, FDM	SLS
material handling	fed continuously	sprayed in layers
wastage	minimum	high
Support	required	may or maynot required
cost	low cost	high cost
post processing	required	not required

	SLS	SLA	LENS
laser Used	CO ₂ laser	UV laser	Nd : YAG laser.
Material	thermoplastic, metal, elastomer	PLA, photopolymer resin.	Inconel, titanium copper, aluminium
Support structure	Not reqd.	Reqd.	Not reqd.
Surface finish	Rough	very Smooth	fairly smooth.

	RP	Conventional.
method	Additive	Subtractive
lead time	Short	Longer.
material usage	Efficient	more wastage.
volume	Low volume	High volume
cost.	low initial cost	High initial cost

Unit 2

1) Solid Ground Curing (SGC)

- The process has two cycles
 - i) Photomask generation cycle
 - ii) Layer fabrication cycle.

Principle

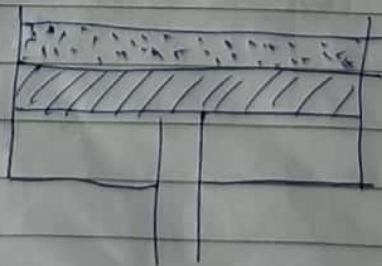
- Parts are built layer by layer from a photosensitive liquid resin that solidifies when exposed to Ultraviolet light.
- The irradiation source is a high powered collimated UV lamp & image of the layer is generated by masked illumination.
- The mask is generated from CAD data and printed on a transparent substrate.

Procedure

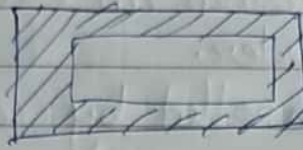
- 1) CAD file is sliced



- 2) A layer of photosensitive liquid resin is sprayed on the platform



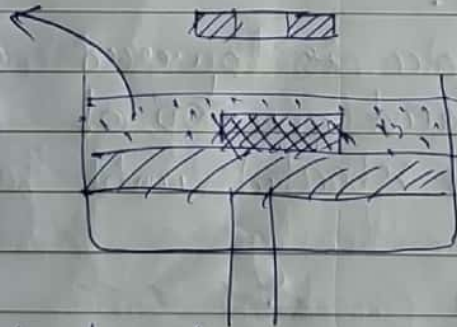
- 3) Photomasks are prepared for each layer using ~~electrography~~ electrophotography.



- 4) The ~~mask~~ mask plate is placed on ^{top of} the build platform and object is under construction.



- 5) UV light is exposed on the layer of polymer. Exposed part solidifies completely.



- 6) The residual polymer is vacuummed for recycling and ~~wax~~ replaced using wax.

- 7) Wax is hardened using wax cooling plate.

- 8) It is milled to accurate thickness for next layer preparation.

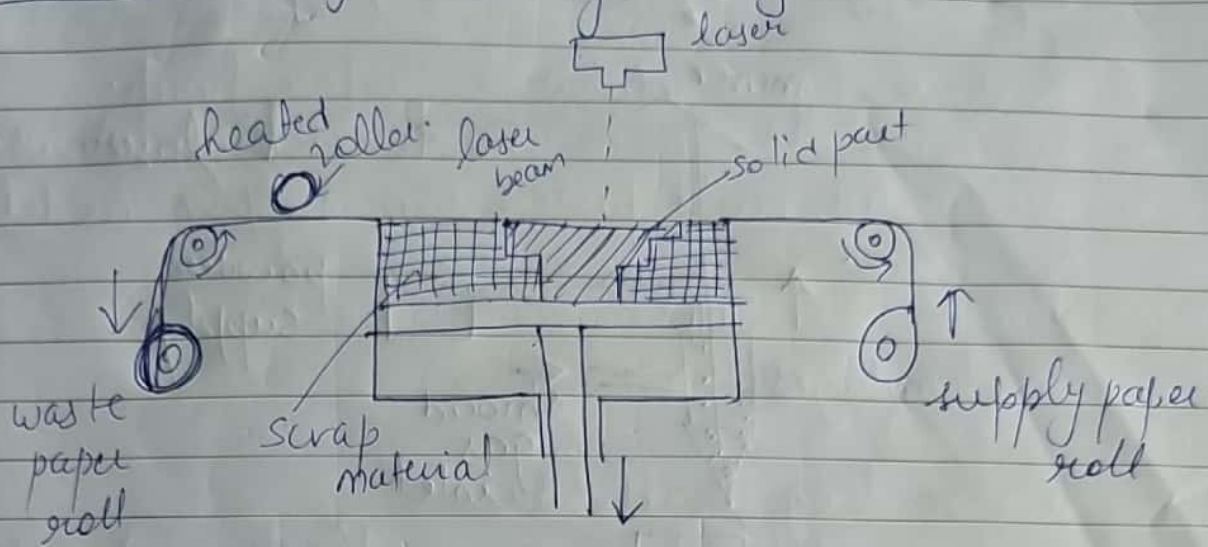
- 9) Mask plate is erased & reused for each layer.

Electron Beam ~~Melting~~ Melting

- Parts are built when an electron beam is fired at the metal powder.
- Electron beam controlled by computer melts the metal powder ~~as~~ in vacuum as per CAD model.
- Building of part is done layer by layer.
- The part to be produced is modelled in CAD program.
- The model is then sliced into thin layers, approximately $1/10^{\text{th}}$ mm.
- A thin layer of powder is scraped on the vertical surface.
- The first layers geometry is created ~~on~~ ^{through} the layer of powder melting as per those points directed by the CAD file. with a computer controlled beam.
- The building surface is lowered and next layer of powder is placed on top of previous layer.
- The procedure is repeated until part is built layer by layer.

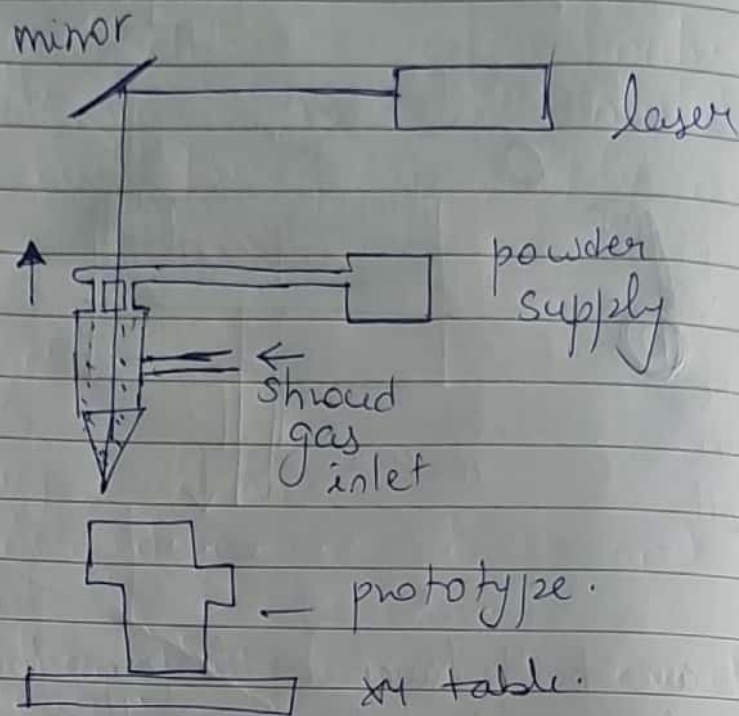
- i) Vacuum chamber
- ii) Electron beam gun
- iii) roller
- iv) monitor
- v) ECU
- vi) Control unit.

Laminated Object Manufacturing (LOM)



- Parts are built layer by layer, by laminating each layer of paper or other sheet like materials and contour of part being cut by CO_2 laser.
- Each layer contains cross section of one or more part.
- The next layer is laminated & built on top of the previous laser cut layer.
- Once the parameters are loaded into LOM Slice, STL file slices as part builds.
- This process of continuous slicing is known as slice on the fly.
- The paper is fed from supply roll to the stack and first bonded to the previous layer using heated roller which melts the plastic coating on the bottom side of the paper.
- The profile is traced by an optic system.
- The laser cuts each profile and roll of paper advances, a new layer is glued onto the stack and process is repeated.

Laser Engineered Net Shaping (LENS)

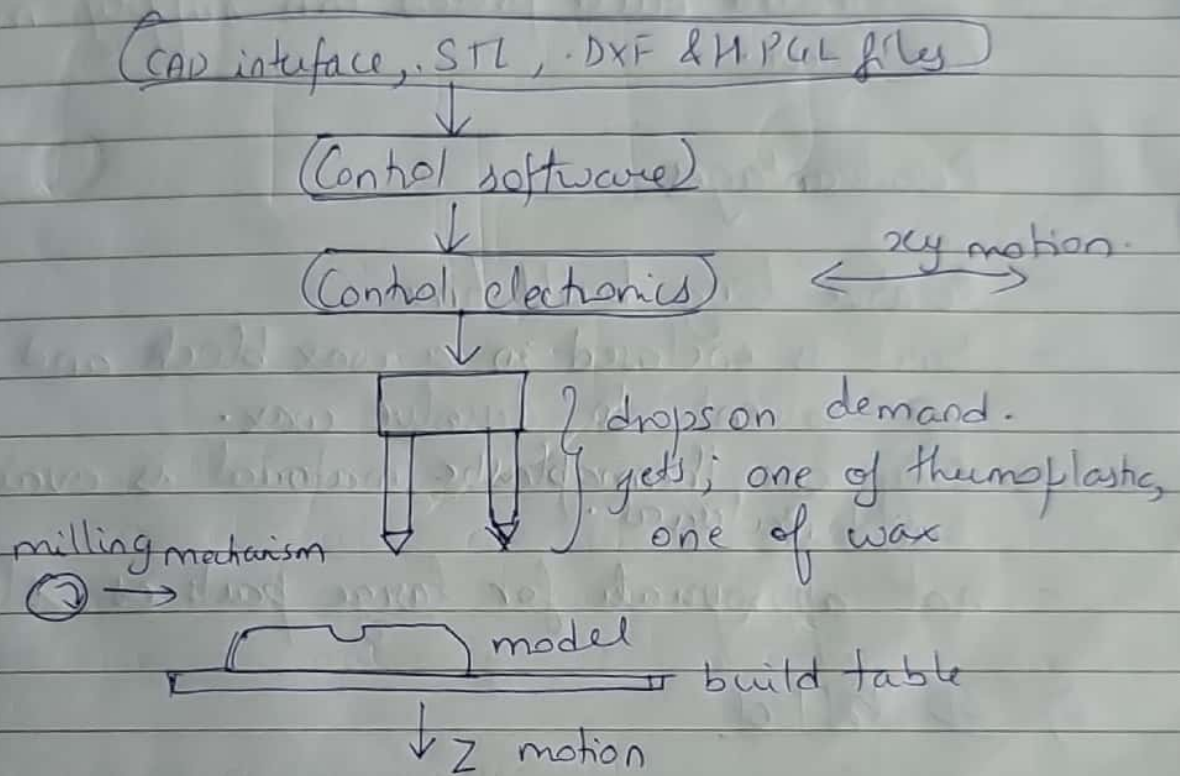


- A high powered Nd:YAG laser focused on a metal substrate creates a molten puddle on substrate surface.
- Powder is injected into the molten puddle to increase material volume.
- A printing motion system moves the platform horizontally & vertically as laser traces the crosssection of the part.
- STL file is sliced & downloaded to the machine bottom slice up.
- A deposition head supplies metal powder to be melted by the Nd:YAG laser.
- Fabrication takes place in low pressure ~~an~~ oxygen chamber for oxygen free operation.
- When the layer is complete, deposition head moves up and continues with next layer.

Unit 3

1) Soft Tooling ————— | ————— Hard Tooling

1) Sanders Model Maker.



- It uses both build and support material to build 3D model.
- Support material is wax based and will melt away leaving only the part, which is made up of build material.
- Material is pumped to the feed line by compressed air within the reservoir.
- There is an actual siphon that is conducted from reservoir to the feed lines, to the printer head during model building.
- The reservoir, feed lines and print head are preheated to allow for continuous flow of material.

Advantages

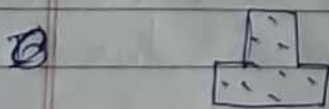
- produces fine resolution & surface finish.
- overcomes the limitations of 2D simulations.
- automated build process can supply models in a few hours.

Disadvantages

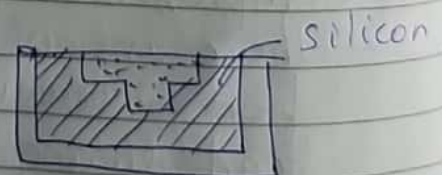
- Relative long build time
- part is enclosed in a wax block and needs ~~to~~ solvent to dissolve wax.
- Only one thermoplastic material is available for build material.
- not appropriate for large parts as slow speed.

2) 3D Keltool

- This process is based on metal sintering process.
- The process starts with a RP pattern from which a rubber mould is created.
- The rubber mould is then used to cast mixture into the mould geometry which after hardening is in green state.
- Green mould is fired and copper infiltrated resulting in a tool.



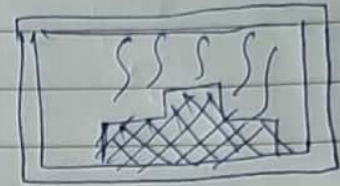
I) master pattern



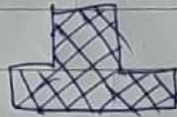
II) silicon casting



III) casting with
tool steel
/ tungsten
carbide/
epoxy mixture



IV) Burnout of binder,
sintering & infiltration
of copper in oven



V) tool insert ready for production.

3) Factors affecting accuracy

i) Factors causing error during data preparation.